

Day 1 Notes

Group Discussion + Aptitude Basics

1) Group Discussion (GD)

Definition: Group Discussion is a round-table discussion where a group of students speak on a given topic for a few minutes. It is commonly used in placements, college activities, interviews, and competitions to evaluate communication, confidence, clarity of thought, leadership, listening ability, and teamwork.

Main purpose of GD:

- To express your ideas clearly
- To listen to others and respond politely
- To present facts, opinions, and examples
- To show confidence, teamwork, and speaking ability

Example GD Topic: "Is social media useful for students?"

Simple way to speak:

"In my opinion, social media is useful for students when it is used in the right way. Students can learn from educational videos, current affairs pages, and career-related content. At the same time, too much use of social media can waste time and reduce concentration. So, social media is useful only when students use it with self-control and for learning purposes."

Tips for GD:

1. Start with a simple definition of the topic.
2. Speak slowly and clearly.
3. Give one or two examples.
4. Do not interrupt others.
5. End with a small conclusion.

2) BODMAS Rule

Definition: BODMAS is the order of operations used to solve arithmetic expressions correctly. It tells us which operation must be performed first when an expression contains brackets, division, multiplication, addition, and subtraction.

BODMAS stands for:

B → Brackets
O → Orders (powers / squares / roots if present)
D → Division
M → Multiplication
A → Addition
S → Subtraction

Example 1: $10 + 2 \times 5$

First do multiplication: $2 \times 5 = 10$

Then add: $10 + 10 = 20$

Example 2: $(12 - 4) \div 2 + 3 \times 2$

Bracket first: $(12 - 4) = 8$

Division: $8 \div 2 = 4$

Multiplication: $3 \times 2 = 6$

Addition: $4 + 6 = 10$

Important point: Multiplication and division are done before addition and subtraction. If multiplication and division both are present, solve from left to right. The same applies to addition and subtraction.

3) Percentage

Definition: Percentage means “per hundred”. It shows a value out of 100. Percentages are widely used in marks, discounts, profit and loss, population changes, and aptitude problems.

Basic formula:

$$\text{Percentage} = (\text{Part} / \text{Total}) \times 100$$

To find a value from percentage:

$$\text{Required value} = (\text{Percentage} / 100) \times \text{Total}$$

Example 1: A student scored 72 marks out of 90. Find the percentage.

$$\text{Percentage} = (72 / 90) \times 100 = 80\%$$

Example 2: Find 25% of 480.

$$= (25 / 100) \times 480$$

$$= 120$$

Shortcut ideas:

50% = half of the number

25% = one-fourth of the number

10% = divide by 10

1% = divide by 100

4) Average

Definition: Average is the central value of a group of numbers. It is obtained by adding all the values and dividing by the number of values.

Formula:

$$\text{Average} = \text{Sum of all observations} / \text{Number of observations}$$

Example 1: Find the average of 10, 20, 30, 40, 50.

$$\text{Sum} = 150$$

$$\text{Average} = 150 / 5 = 30$$

Example 2: The marks of 4 subjects are 70, 80, 90, 60. Find the average marks.

$$\text{Sum} = 300$$

$$\text{Average} = 300 / 4 = 75$$

Useful reverse formula:

$$\text{Total} = \text{Average} \times \text{Number of observations}$$

5) Ratio

Definition: A ratio compares two quantities of the same kind using division. It shows how many times one quantity contains another quantity. Ratios are usually written with a colon (:).

General form:

$$a : b$$

This means a / b

Simplifying a ratio:

Divide both terms by their common factor.

Example 1: Simplify 20 : 30

$$\text{Common factor} = 10$$

$$20 : 30 = 2 : 3$$

Example 2: The ratio of boys to girls is 18 : 12.
Simplified ratio = 3 : 2

Important rule: Both quantities must be in the same units before forming a ratio.

6) Finding Ratio When Two Values Are Given

Definition: If two quantities are given directly, we can write them in ratio form and simplify it.

Steps:

1. Write the quantities in the form $a : b$
2. Find the HCF / common factor
3. Divide both terms by that factor

Example 1: Find the ratio of 45 and 60.

Ratio = 45 : 60

Divide both by 15

= 3 : 4

Example 2: Find the ratio of 250 ml and 1 litre.

Convert to same units first: 1 litre = 1000 ml

Ratio = 250 : 1000 = 1 : 4

Answer pattern: Always reduce the ratio to its simplest form.

7) Total Given, Find the Actual Values from Ratio

Definition: When the ratio of two quantities and their total are given, we can find the actual values by first finding the sum of ratio parts.

Method:

If the ratio is $a : b$ and total is T ,

Sum of ratio parts = $a + b$

First value = $(a / (a+b)) \times T$

Second value = $(b / (a+b)) \times T$

Example 1: The ratio of two numbers is 2 : 3 and their total is 40. Find the numbers.

Sum of ratio parts = 5

First number = $(2/5) \times 40 = 16$

Second number = $(3/5) \times 40 = 24$

Example 2: The ratio of boys and girls is 4 : 5. Total students = 54. Find boys and girls.

Sum of parts = 9

Boys = $(4/9) \times 54 = 24$

Girls = $(5/9) \times 54 = 30$

Shortcut: Divide the total by the sum of ratio parts. Then multiply by each ratio part.

Example 2: $54 \div 9 = 6$, so boys = $4 \times 6 = 24$ and girls = $5 \times 6 = 30$.

8) If One Quantity and Ratio Are Given, Find the Other Quantity

Definition: Sometimes one part of the ratio is known. In that case, first find the multiplication factor and then find the other part.

Method:

If ratio is $a : b$ and first quantity = known value,

Factor = known value / a

Second quantity = $b \times$ factor

Example: The ratio of two numbers is 3 : 5. If the first number is 18, find the second number.
Ratio part 3 corresponds to 18
Factor = $18 \div 3 = 6$
Second number = $5 \times 6 = 30$

So the two numbers are 18 and 30.

9) Mixed Practice Examples

Below are a few mixed aptitude examples combining the concepts learned today.

1. BODMAS: $20 - 4 \times 3 + 8 \div 2$
 $= 20 - 12 + 4 = 12$

2. Percentage: Find 15% of 200
 $= (15/100) \times 200 = 30$

3. Average: Find average of 8, 12, 16, 20
Sum = 56, Average = $56/4 = 14$

4. Ratio: Divide 72 in the ratio 5 : 7
Sum of parts = 12
One part = $72/12 = 6$
Required numbers = 30 and 42

These are standard aptitude patterns that should be practiced repeatedly for speed and accuracy.

10) Quick Summary

Today's topics:

- Group Discussion basics and speaking tips
- BODMAS order of operations
- Percentage formulas and examples
- Average formula and reverse formula
- Ratio basics and simplification
- Finding ratio from two numbers
- Dividing total based on ratio
- Finding unknown quantity from ratio

Remember:

1. In BODMAS, follow the order correctly.
2. In percentage, percent means out of 100.
3. In average, add all values and divide by count.
4. In ratio, always simplify and convert to the same units first.
5. When total and ratio are given, use the sum of ratio parts.